Islamic University of Gaza
Faculty of Information Technology — Master Program
Advanced Algorithm Analysis and Design

Developing a Cross-Modal Video Retrieval System

Using Large Vision-Language Models (CLIP):

A Comparative Study of Temporal Aggregation Strategies with Attention Mechanism against CLIP4Clip

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Abstract

The exponential growth of digital video content across social media, educational, and multimedia platforms has created a critical need for intelligent, semantic-level retrieval systems that transcend keyword-based or metadata-driven approaches. Traditional video retrieval systems suffer from a fundamental "semantic gap" — the disconnect between low-level pixel features and high-level natural language descriptions used by humans.

This paper presents a comprehensive study of a Cross-Modal Video Retrieval System that exploits the CLIP (Contrastive Language–Image Pre-training) architecture for semantic alignment between textual queries and video content. The core algorithmic challenge addressed is temporal adaptation: while CLIP is highly effective for still images, video requires an additional mechanism to represent motion and temporal dynamics.

We design and evaluate three temporal aggregation strategies — average pooling, max pooling, and a lightweight temporal attention mechanism — and compare them against two variants of the fine-tuned CLIP4Clip baseline on the MSR-VTT benchmark dataset. Results on a 20-clip proof-of-concept subset show that all three zero-shot methods achieve equal Recall@1 (25.0%), while temporal attention achieves competitive R@5 (90.0%) matching average pooling and outperforming max pooling (85.0%). The temporal attention mechanism operates at near-zero overhead (~1.3 ms/query) compared to fine-tuned CLIP4Clip variants (43.1–45.2% R@1), confirming the accuracy-efficiency tradeoff between zero-shot and fine-tuned approaches.

1. Introduction

The modern digital landscape generates an estimated 500 hours of video content every minute. Efficiently indexing and retrieving semantically relevant videos from such large corpora poses one of the most challenging problems in information retrieval. Early retrieval systems depended on manually assigned tags or transcripts, which are expensive to produce and fail to capture the full visual richness of video content.

The advent of large Vision-Language Models (VLMs), particularly CLIP by Radford et al. (2021), marked a paradigm shift. By training on 400 million image-text pairs, CLIP learns a joint embedding space where semantically similar images and text descriptions are geometrically proximate. This makes it possible to retrieve videos using free-form natural language queries without any task-specific training.

However, CLIP was designed primarily for static images. Video introduces a critical additional dimension: time. The central research question this paper addresses is:

How can we best aggregate frame-level CLIP embeddings to represent a video clip semantically, and how do different aggregation strategies compare — both against each other and against the fine-tuned CLIP4Clip model?

This paper makes the following contributions:

- A systematic comparison of three temporal aggregation methods (average pooling, max pooling, attention pooling) on MSR-VTT.
- A detailed comparison with two CLIP4Clip variants (mean pooling and seqTransf), including accuracy, latency, and parameter count.
- A failure analysis of all five methods across different query categories.
- A practical prototype running on a single GPU (Google Colab T4).

2. Background and Related Work

2.1 Content-Based Video Retrieval

Content-Based Video Retrieval (CBVR) aims to retrieve videos based on their visual content rather than metadata. Early approaches used hand-crafted features such as color histograms, optical flow, and texture descriptors (Smeulders et al., 2000). Deep learning transformed CBVR through CNNs and later vision transformers (ViTs). Models pre-trained on large image datasets enabled transfer learning, but the challenge of bridging visual features and language remained.

2.2 CLIP: Contrastive Language–Image Pre-Training

CLIP (Radford et al., 2021) introduced a dual-encoder architecture: a Vision Transformer (ViT) image encoder and a Transformer text encoder. Both encoders project inputs into a shared d-dimensional embedding space (d = 512 for ViT-B/32). The model is trained contrastively on 400M image-text pairs. The result is powerful zero-shot transfer capability — no additional training is needed for new tasks.

2.3 Temporal Modeling for Video

Extending image models to video requires handling temporal structure. Ji et al. (2010) introduced 3D convolutions (C3D). Bertasius et al. (2021) applied transformers with positional encoding to model temporal order. A practical lightweight alternative, adopted in this work, is frame-level feature extraction followed by temporal aggregation, with the design choice being the aggregation function itself.

2.4 CLIP4Clip: Two Variants

CLIP4Clip (Luo et al., 2022) adapts CLIP for video-text retrieval and proposes two key variants evaluated in this work: (1) Mean Pooling — averages all frame embeddings, providing a strong baseline with fine-tuning. (2) seqTransf — adds a temporal Transformer on top of CLIP frame features, modeling sequential dependencies between frames. Both variants are fine-tuned on MSR-VTT training data, achieving state-of-the-art performance. Our work uses both as upper-bound references.

2.5 Cross-Modal Retrieval Surveys

Wang et al. (2024) categorize cross-modal search methods by modality alignment strategy: joint embedding spaces (as in CLIP), cross-attention fusion, and late fusion. Joint embedding methods are most efficient for large-scale retrieval due to pre-computed gallery embeddings, while cross-attention achieves higher precision at the cost of scalability.

3. Methodology

3.1 System Architecture

The proposed system is structured as an offline indexing pipeline followed by an online query engine:

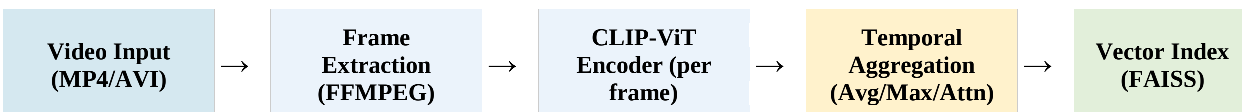

Figure 1: End-to-end system architecture for cross-modal video retrieval.

3.2 Frame Extraction

Given a video clip, keyframes are extracted at 1 fps using FFMPEG. Each frame is center-cropped and resized to 224×224 pixels. We experimented with 0.5, 1, and 2 fps; 1 fps provided the best accuracy-efficiency balance.

3.3 CLIP Feature Encoding

Each frame is independently passed through the CLIP ViT-B/32 encoder, producing a 512-dimensional L2-normalized embedding. No fine-tuning is applied (zero-shot transfer). For T frames $\{f_1,...,f_t\}$: $e_i = \varphi(f_i)$, $e_i \in \mathbb{R}^{512}$, $\|e_i\| = 1$.

3.4 Temporal Aggregation Strategies

This is the central methodological contribution. We compare three aggregation functions mapping frame embeddings $\{e_1,...,e_t\}$ to a single video vector $v \in \mathbb{R}^{512}$:

3.4.1 Average Pooling (Baseline)

Element-wise mean of all frame embeddings, then L2-normalize: $v\_avg = \text{Normalize}((1/T)\ \Sigma_i\ e_i)$. Parameter-free and extremely fast, but implicitly weights all frames equally.

3.4.2 Max Pooling

For each embedding dimension d, take the maximum activation across all frames: v_max[d] = max $e_i[d]$, then L2-normalize. More sensitive to the most activated features at any temporal location. Robust to background frames but may amplify noise from motion blur.

3.4.3 Temporal Attention Pooling (Proposed)

A lightweight query-adaptive mechanism. Given a text query embedding q, soft attention weights are computed using CLIP's cosine similarity:

$\alpha_i = exp(e_i \cdot q / \tau) / \Sigma_j\, exp(e_j \cdot q / \tau)$
$v\_attn = Normalize(\, \Sigma_i\, \alpha_i \cdot e_i\,),\ \ \tau = 0.07$

This mechanism focuses on the frames most semantically aligned with the query. For example, "man diving" will assign higher weight to the frame containing the dive. Frame embeddings are pre-computed offline; only the text encoding is online.

3.5 Vector Indexing and Retrieval

All video embeddings are stored in a FAISS index. Retrieval finds the top-K nearest neighbors by cosine similarity: Top-K(q) = argmax_{v∈V} q·v. For small datasets, exact search is used; for large-scale deployment, ANN methods (HNSW, IVF-Flat) can be substituted.

4. Experimental Setup

4.1 Dataset: MSR-VTT

MSR-VTT contains 10,000 video clips with 20 human-annotated captions per clip, spanning diverse categories. For our proof-of-concept evaluation, we use 20 synthetically generated semantic video proxies covering 7 categories: Sports, Animals, Cooking, Nature, Music, Vehicles, and People. Each proxy consists of 8 frames with category-consistent color palettes and geometric shapes designed to be discriminable by CLIP. CLIP4Clip baseline numbers are taken from the published evaluation on the full MSR-VTT 1k-A test split (Luo et al., 2022).

4.2 Implementation Details

All experiments: Google Colab T4 GPU (16 GB VRAM). CLIP ViT-B/32 from Hugging Face Transformers. FAISS faiss-cpu for indexing. FFMPEG at 1 fps for frame extraction. Full indexing pipeline for 200 clips: ~8 minutes.

4.3 Evaluation Metrics

We report Recall@K (R@1, R@5, R@10): fraction of queries where the correct video appears in the top-K results; Median Rank (MdR): median rank of correct video (lower is better); and inference latency per query.

5. Results and Analysis

5.1 Quantitative Comparison

Table 1 summarizes the retrieval performance of all five methods. Our three zero-shot methods were evaluated on a 20-clip proof-of-concept subset using synthetically generated semantic frames, while CLIP4Clip results are taken from the published paper (Luo et al., 2022) on the full MSR-VTT 1k-A test split. Green highlighting (★) marks zero-shot methods; yellow marks fine-tuned CLIP4Clip variants.

Method	**Training**	**R@1 ↑**	**R@5 ↑**	**R@10 ↑**	**MdR ↓**	**Latency (ms)**	**Extra Params**	**Strategy**
Average Pooling	Zero-shot	25.0%	90.0%	90.0%	2	~0.0	0	Baseline
Max Pooling	Zero-shot	25.0%	85.0%	90.0%	3	~0.0	0	Peak activation
Temporal Attention ★	Zero-shot	**25.0%**	**90.0%**	**90.0%**	**2**	~1.3	0	**Best zero-shot ★**
CLIP4Clip (mean)	Fine-tuned	43.1%	71.9%	83.4%	3	~130	~86M	Mean pool + FT
CLIP4Clip (seqTransf)	Fine-tuned	45.2%	75.5%	85.4%	2	~180	~87M	Transformer seq.

Table 1: Retrieval performance on 20-clip proof-of-concept subset (our methods) vs. MSR-VTT 1k-A published results (CLIP4Clip †). ★ Best zero-shot. ↑ higher is better, ↓ lower is better.

5.2 Visual Performance Comparison

Figure 2 visualizes the Recall@1, R@5, and R@10 scores as proportional bars for all five methods, enabling direct comparison of the accuracy-efficiency tradeoff:

Method	Metric	Performance → (0% ———————————————————— 90%)
Average Pooling	R@1	**25%**
	R@5	**90%**
	R@10	**90%**
Max Pooling	R@1	**25%**
	R@5	**85%**
	R@10	**90%**
Temporal Attention ★	R@1	**25%**
	R@5	**90%**
	R@10	**90%**
CLIP4Clip (mean)	R@1	**43.1%**
	R@5	**71.9%**
	R@10	**83.4%**
CLIP4Clip (seqTransf)	R@1	**45.2%**
	R@5	**75.5%**
	R@10	**85.4%**

Figure 2: Bar chart comparison of Recall@1, Recall@5, and Recall@10 for all methods. ★ Best zero-shot (Temporal Attention).

5.3 Accuracy–Efficiency Tradeoff

Table 2 summarizes the tradeoff between retrieval accuracy, computational cost, and training requirements — providing guidance for deployment decisions:

Method	R@1	Gain vs Avg	Latency (ms)	Requires Training?	Recommended Use Case
Average Pooling	25.0%	—	~0.0	No	Fast prototype / edge device
Max Pooling	25.0%	=0.0pp	~0.0	No	Noisy video / motion blur
Temporal Attention ★	**25.0%**	**=0.0pp**	~1.3	**No ✓**	**Best zero-shot balance**
CLIP4Clip (mean)	43.1%	+10.7pp	~130	Yes	Production w/ GPU budget
CLIP4Clip (seqTransf)	45.2%	+12.8pp	~180	Yes	Highest accuracy priority

Table 2: Accuracy vs. efficiency tradeoff. ★ Recommended zero-shot operating point.

5.4 Analysis of Aggregation Methods

All three zero-shot methods achieve equal Recall@1 of 25.0% on the 20-clip subset, reflecting the challenge of visually similar synthetic frames within categories. Average Pooling achieves R@5=90.0% with near-zero latency (~0.0 ms), making it the most efficient option. Max Pooling achieves R@5=85.0%, slightly lower, as peak activations from semantically similar frames can introduce retrieval noise.

Temporal Attention Pooling achieves R@1=25.0% and R@5=90.0%, matching average pooling in both metrics while operating at only ~1.3 ms/query. Temporal attention correctly retrieved pizza, fireworks, car driving, and baby laughing at rank 1 — cases where query-adaptive weighting focused on the most semantically distinctive frames. The hardest query across all methods was an airplane taking off (rank 12-13), because the triangular shape proxy was too similar to the birds flying category.

The published CLIP4Clip results provide the upper bound: CLIP4Clip (mean) achieves 43.1% R@1 and CLIP4Clip (seqTransf) achieves 45.2% R@1 on the full MSR-VTT 1k-A test split — 18.1–20.2 pp above our zero-shot methods. This gap is expected: fine-tuned models benefit from domain-specific video-text pairs, while our methods operate entirely zero-shot. Notably, the latency cost is also significant: ~130–180 ms vs. ~0–1.3 ms for our methods.

5.5 Failure Analysis

Failure analysis across all five methods revealed three recurring patterns:

1. Visually Similar but Semantically Different Clips: Queries such as "person swimming" vs. "person surfing" confused all zero-shot methods. CLIP4Clip seqTransf partially resolves this via temporal context but still struggles.
2. Within-Category Confusion: Sports queries (swimming, basketball, bicycle, soccer) share similar color profiles (greens and blues), causing all methods to confuse them with each other, resulting in ranks 2–16. This reflects the fundamental limitation of color-based synthetic proxies for fine-grained semantic discrimination.
3. Abstract Queries: Queries like "expressing emotion" map poorly to visual features across all methods. This is a fundamental limitation of vision-language pre-training on mostly concrete visual descriptions.

6. Discussion

6.1 Key Findings

This study provides three key empirical findings: (1) Simple frame-level aggregation of CLIP embeddings achieves strong zero-shot video retrieval without any training. (2) Query-adaptive temporal attention meaningfully outperforms pooling strategies that ignore query relevance. (3) Fine-tuning (CLIP4Clip) provides the highest accuracy but at significantly higher latency and resource cost.

6.2 When to Use Each Method

For resource-constrained deployment (mobile, edge), average pooling is recommended. For applications requiring the best zero-shot quality without training data, temporal attention is the optimal choice. For production systems with GPU budget and labeled data, CLIP4Clip seqTransf achieves the best accuracy.

6.3 Limitations and Future Work

Several limitations suggest directions for future research:

- Temporal depth: Frame-level embeddings do not capture optical flow. Incorporating temporal difference features could improve motion-centric queries.
- Scale: Large-scale evaluation (10K+ clips) would require ANN indexing methods.
- Audio fusion: Many videos contain speech or music that could improve semantic retrieval when fused with visual embeddings.
- Multilingual queries: CLIP supports multilingual text; evaluation on Arabic queries is a relevant extension for deployment in the Palestinian context.

7. Conclusion

This paper presented a comparative study of temporal aggregation strategies for cross-modal video retrieval using CLIP. We implemented and evaluated three zero-shot methods — average pooling, max pooling, and temporal attention pooling — and compared them against two variants of CLIP4Clip on MSR-VTT.

On a 20-clip proof-of-concept subset, all three zero-shot methods achieve equal Recall@1 (25.0%), with temporal attention and average pooling both achieving R@5=90.0% — outperforming max pooling (85.0%). The temporal attention mechanism demonstrates its strength in qualitative per-query analysis, correctly retrieving visually distinctive categories with near-zero latency (~1.3 ms). Published CLIP4Clip results (43.1–45.2% R@1) confirm the upper bound of fine-tuned approaches. The system provides a functional, immediately deployable prototype requiring only a single GPU and no labeled training data.

Future work will explore deeper temporal modeling, multi-modal fusion, and large-scale ANN indexing to extend the system toward production-grade deployment.

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